Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

_____show ip route ospf ____

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? ____S0/0/0________

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? ___129______

Does R2 show up as an OSPF neighbor on R1? ___Yes_____

Does R2 show up as an OSPF neighbor on R3? ____No____

What does this information tell you?

___With all interfaces on R2 set to passive by default, R2 will not send any message on the interfaces, therefore OSPF neighbor adjacencies cannot form, which lead to R3 missing the neighbor relationships with R2.___________

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h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

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R2 (config) # router ospf 1

R2 (config-router) # no passive-interface s0/0/1

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i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? ____S0/0/1________

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What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

_____65_____ 2

Is R2 listed as an OSPF neighbor to R3? __Yes_______ 2

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
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Why would you want to change the OSPF default reference-bandwidth?

_____Because we want to obtain a more accurate cost calculation for higher-speed link 2

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Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

__192.168.3.0/24 (cost=782): R1 S0/0/1 (cost=781) + R3 G0/0 (cost=1)

__192.168.23.0/30 (cost=845) : R1 S0/0/1 (cost=781) + R3 S0/0/1 (cost=64) 2

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i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

_1562. Because both serial links now have the same bandwidth (128 Kbps), hence the cost for each serial link is the same (i.e. 781+781=1562) ____________________________ 2

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Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

_OSPF always chooses the path with the lowest cost, hence R1 now travel via R2. 4

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   __Because if the router ID is not controlled, OSPF automatically selects the highest IP address on loopback interfaces, which can lead to unpredictable router IDs. Therefore, controlling router ID can ensure consistency. 4

2. Why is the DR/BDR election process not a concern in this lab?

   Because the DR/BDR election only occurs on multi-access networks (e.g. Ethernet). In this lab, all serial links are point-to-point connections, which do not require DR/BDR election. 4

3. Why would you want to set an OSPF interface to passive?

   __It can help prevent OSPF from sending useless information out of interface. 4

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   __OSPF's use of areas. Area 0 (i.e. the backbone area) connects to all other areas, limiting routing updates and isolating topology change.___ 4

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   _network 10.0.0.0 0.255.255.255 area 0 4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   EIGRP (4039043) will be used 4